

Research on Weed Instance Segmentation and Maize Row-Band Detection

Internship Summary Report

Internship Duration: Eight Weeks

Datasets: CropAndWeed / Maize Row-Band Dataset

Research Directions: Weed Instance Segmentation and Crop Row-Band Detection

1. Project Background and Objectives

In precision agriculture, weed recognition, target localization and crop-row detection are key perception components for autonomous weeding and navigation by intelligent agricultural machinery. Manual weeding is costly and inefficient, while image-based automatic detection methods can provide more stable and quantifiable target information in real field environments. This internship focused on two experimental tasks using real farmland images: six-class weed instance segmentation and maize seedling row-band detection.

The project consisted of two main objectives. The first was to conduct pixel-level instance segmentation experiments for six common weed classes: cockspur_grass, black_nightshade, field_milk_thistle, meadow_grass, redroot_amaranth and white_goosefoot. The second was to implement maize row-band detection based on image segmentation algorithms, with emphasis on both detection accuracy and real-time performance.

Across the eight weeks, the work covered fundamental digital image processing, dataset screening and annotation, experiments with SAM/SAM2 prompt-based models, Mask2Former end-to-end instance segmentation, Grounded SAM2 automatic segmentation, and BiSeNet-V2 training and post-processing for maize row-band detection.

Overall Experimental Workflow

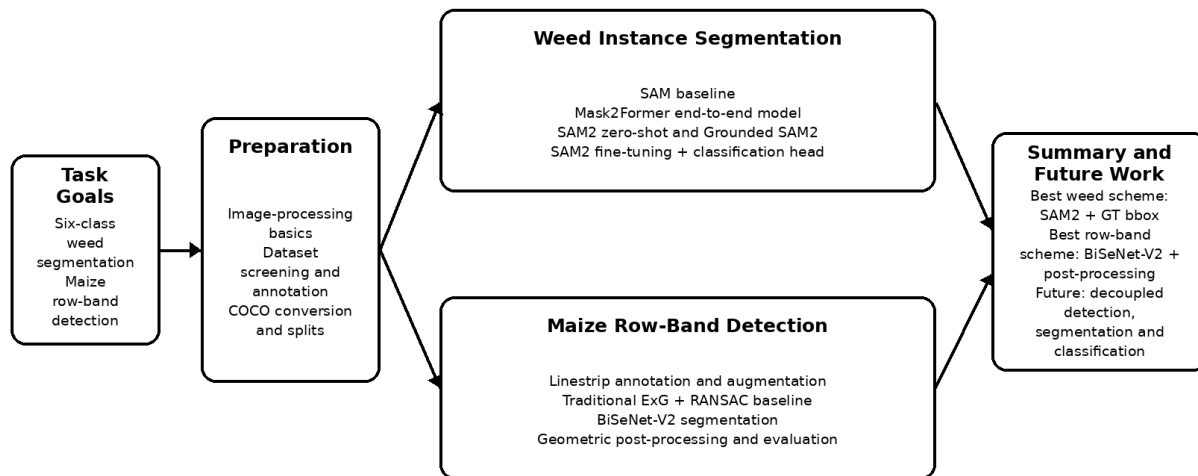


Figure 1. Overall experimental workflow

2. Overall Results

For weed instance segmentation, SAM2 zero-shot inference with GT bbox prompts achieved the best result in this project. The AP@0.5 on the test set reached 63.0%, and the AP@0.5 values of four broadleaf weed classes exceeded 90%. However, grass-type weeds remained the main bottleneck because their targets were thin, small in area and occupied only a small proportion of the bounding box.

For fully automatic end-to-end instance segmentation, Mask2Former achieved only 11.6% AP@0.5 after dataset expansion, while Grounded SAM2 achieved only 0.2% AP@0.5 because the front-end Grounding DINO detector struggled to detect extremely small targets. This indicates that the main limitation of the automatic weed segmentation pipeline was not a single model architecture, but the combined difficulty of extremely small targets, complex farmland backgrounds and similar class appearances.

For maize row-band detection, BiSeNet-V2 combined with geometric post-processing reduced the mean angle error from 18.14 degrees in the traditional baseline to 2.14 degrees, and reduced the mean distance error from 70.6 px to 8.6 px. The inference speed reached 56.5 fps, satisfying the real-time requirement of more than 20 fps. The main limitation was that the detection rate was only 58.2%, with most missed detections occurring in sparse row bands and edge-cropped samples.

Task	Best Current Scheme	Main Results	Main Limitation
Weed instance segmentation	SAM2 zero-shot + GT bbox prompt	Overall AP@0.5 = 63.0%; AP@0.5 > 90% for four broadleaf classes	Still depends on bbox prompts; weak performance for grass-type weeds
Fully automatic weed segmentation	Mask2Former / Grounded SAM2	Mask2Former v3 AP@0.5 = 11.6%; Grounded SAM2 AP@0.5 = 0.2%	Extremely small targets are difficult to detect; DINO stage fails obviously
Maize row-band detection	BiSeNet-V2 + column projection + RANSAC	Angle error = 2.14 degrees; distance error = 8.6 px; speed = 56.5 fps	Detection rate = 58.2%; sparse and edge-cropped row bands are often missed

Table 1. Overview of overall experimental results

3. Weekly Work Summary

Week 1: Learned the fundamentals of digital image processing, including grayscale transformation, histogram equalization, spatial filtering, edge detection, threshold segmentation and morphology. OpenCV experiments helped establish the basic concepts needed for later segmentation work.

Week 2: Screened weed datasets and selected CropAndWeed as the main data source for six weed classes. Web-crawled images were rejected because of watermarks, non-field backgrounds and large style differences, and manual polygon annotation plus data augmentation was selected as the main dataset-building strategy.

Week 3: Built a SAM vit_l zero-shot baseline using bbox prompts and tested decoder-only fine-tuning. The model performed well on broadleaf weeds but showed clear limitations on thin grass-type targets.

Week 4: Attempted to reproduce SeqFormer and ViT-CoMer, but both were limited by old dependencies and compatibility problems in current cloud environments. Mask2Former was therefore selected as a practical end-to-end instance segmentation alternative.

Week 5: Expanded the dataset to coco_v3 and retrained Mask2Former, but the AP improvement was limited because many weed instances were extremely small. SAM2 zero-shot inference and Grounded SAM2 were then evaluated, showing that SAM2 had strong segmentation ability with GT bbox prompts while Grounding DINO was ineffective for automatic small-target detection.

Week 6: Built the SAM2 fine-tuning + classification head framework and began diagnostic training. In parallel, completed maize row-band annotation, offline data augmentation and a traditional ExG + projection + RANSAC baseline.

Week 7: Converted linestrip annotations to binary masks, trained BiSeNet-V2 for maize row-band segmentation, and implemented the geometric post-processing pipeline. The method significantly improved angle error, distance error and speed compared with the traditional baseline.

Week 8: Completed the final SAM2 fine-tuning + classification head evaluation and fixed key issues related to bbox coordinate transformation and category evaluation. The results showed that mask decoder fine-tuning was partially effective, but joint classification did not converge reliably, so decoupled segmentation and classification is recommended for future work.

4. Preliminary Study and Dataset Construction

4.1 Fundamental Digital Image Processing Study

The first week focused on learning basic digital image processing methods, including image representation, grayscale transformation, histogram equalization, spatial filtering, edge detection, threshold segmentation and mathematical morphology. Corresponding experiments were implemented with OpenCV. This stage helped establish the basic concepts of image segmentation and clarified the roles of target enhancement, noise suppression and foreground segmentation in a typical image-processing workflow.

The experiments showed that median filtering was better than mean filtering and Gaussian filtering for salt-and-pepper noise. Canny edge detection produced clear edges when the thresholds were properly selected. Otsu thresholding could automatically determine a binary threshold, but it was difficult to obtain stable segmentation results in complex farmland images with shadows and illumination changes. These limitations of traditional methods further explain the necessity of deep learning methods for complex segmentation tasks.

4.2 Weed Dataset Screening and Class Selection

During dataset construction, public datasets and web-crawled images were screened according to three requirements: real farmland background, six common weed classes and suitability for instance segmentation. CropAndWeed was finally selected as the main data source because its images were collected from real overhead farmland scenes with complex backgrounds, which better matched the project application scenario.

For data supplementation, Baidu image crawling was tested, but the collected images contained watermarks, white backgrounds, close-up shots, non-farmland scenes and excessive style differences. Directly adding these images would have damaged the dataset distribution. Therefore, the final strategy was not to force the sample count with web images, but to supplement data through bbox-expanded cropping and offline data augmentation.

Class	Original Images	Proportion of Extremely Small Targets
cockspur_grass	266	28.6%
black_nightshade	400	96.2%
field_milk_thistle	470	73.6%
meadow_grass	500	94.4%
redroot_amaranth	470	60.0%
white_goosefoot	500	69.2%

Table 2. Statistics of selected original images for six weed classes

4.3 Annotation Method and Quality Control

In terms of annotation tools, edgeSAM was first tested for semi-automatic segmentation to improve efficiency. However, for extremely small overhead farmland targets, edgeSAM often incorrectly segmented soil, shadows or nearby green regions as the target, and the boundary quality was poor. Therefore, the formal dataset mainly used manual polygon annotation to ensure usable instance-level masks.

Considering that some targets were very small, each image could contain many instances, and grass-type leaves were thin and elongated, the samples were divided into A/B/C quality groups during dataset construction. Group A contained clear targets with high-quality annotations; Group B contained difficult but still useful training samples; Group C mainly contained small-target supplementary samples obtained by bbox-expanded cropping.

Dataset	Images	Instances	Description
Training set	2571	8687	A/B + C crops

Dataset	Images	Instances	Description
Validation set	180	574	A/B, no augmentation
Test set	180	544	A/B, no augmentation

Table 3. coco_v3 dataset split

5. Weed Instance Segmentation Experiments

5.1 SAM Zero-Shot and Decoder-Only Fine-Tuning

In the third week, SAM vit_l was first used as the segmentation baseline with official pretrained weights. Bbox prompts were used for zero-shot inference. A combined bbox + stem-point prompt was also tested, but the overall result decreased, so the formal baseline used bbox-only prompts.

The SAM zero-shot results showed good performance on broadleaf weeds. The mean IoU values of black_nightshade, field_milk_thistle, redroot_amaranth and white_goosefoot were all above 0.77. However, the model was weaker on grass-type thin-leaf targets such as cockspur_grass and meadow_grass. The main reason was that grass-type leaves were thin and occupied only a small part of the bbox, so the bbox prompt did not provide enough information.

Class	Instances	Mean IoU	IoU ≥ 0.5	IoU ≥ 0.9
black_nightshade	141	0.7666	99.3%	8.5%
cockspur_grass	708	0.5793	71.3%	0.4%
field_milk_thistle	212	0.7713	98.1%	12.7%
meadow_grass	192	0.5192	60.9%	0.0%
redroot_amaranth	177	0.8360	96.0%	33.9%
white_goosefoot	141	0.8472	97.9%	32.6%
overall	1571	0.6676	81.3%	9.4%

Table 4. SAM zero-shot experimental results

Because of GPU memory limits, the subsequent fine-tuning experiment froze the image encoder and prompt encoder and trained only the mask decoder. After fine-tuning, the four broadleaf classes improved to some extent, but the overall IoU decreased from 0.6676 to 0.6289. This shows that decoder-only fine-tuning can improve mask details for some classes, but it provides limited help for thin grass-type targets and is less stable overall than the zero-shot baseline.

Class	FT IoU	ZS IoU	Change	IoU ≥ 0.5
cockspur_grass	0.5311	0.5790	-0.0479	0.670
black_nightshade	0.7900	0.7670	+0.0230	1.000
field_milk_thistle	0.8181	0.7710	+0.0471	1.000
meadow_grass	0.5282	0.5190	+0.0092	0.667
redroot_amaranth	0.8460	0.8360	+0.0100	1.000
white_goosefoot	0.8268	0.8470	-0.0202	1.000
overall	0.6289	0.6676	-0.0387	0.780

Table 5. SAM decoder-only fine-tuning results

5.2 SeqFormer, ViT-CoMer and Mask2Former

According to the internship plan, the fourth week first attempted to reproduce SeqFormer and ViT-CoMer. However, both official implementations were based on older environments and were incompatible with the current cloud platform. Mask2Former was therefore selected as an end-to-end instance segmentation alternative with better compatibility. It supports COCO format and can directly use the converted dataset under Python 3.12 + PyTorch 2.x. The experiments expanded from a 500-image A/B dataset to the 2571-image coco_v3 training set, but AP@0.5 was still only 11.6%.

Version	Training Data	Main Configuration	AP@0.5
v1	500 images	All-zero masks due to annotation-reading bug	Invalid
v2	coco_v2	batch = 4, 640 x 800, 18 epochs	12.0%
v3	coco_v3	Data augmentation + early stopping, triggered at epoch 11	11.6%

Table 6. Mask2Former experimental results

After dataset expansion, Mask2Former did not improve noticeably. The main reason was that the median instance area in the dataset was only about 0.05%-0.07%, and many targets became nearly invisible after image resizing during training. Meanwhile, the farmland background was complex and the classes were visually similar, making it difficult for an end-to-end model to learn sufficiently under the current data scale and resolution constraints.

5.3 SAM2 Zero-Shot and Grounded SAM2 Automatic Experiments

Because Mask2Former did not perform well, SAM2 vit_l was introduced for zero-shot inference. The experiment used GT bboxes from the coco_v3 test set as prompts, without any training, and used COCO AP@0.5 for a unified comparison with Mask2Former.

Class	SAM2 Zero-Shot AP@0.5	Mask2Former v3 AP@0.5
cockspur_grass	0.487	0.211
black_nightshade	0.913	0.087
field_milk_thistle	0.941	0.167
meadow_grass	0.342	0.041
redroot_amaranth	1.000	0.102
white_goosefoot	0.960	0.335
overall	0.630	0.116

Table 7. Comparison between SAM2 zero-shot and Mask2Former test results

The results show that SAM2 zero-shot inference achieved an overall AP@0.5 of 63.0%, much higher than the 11.6% of Mask2Former v3. Four broadleaf weed classes all exceeded 90% AP@0.5, indicating that the general segmentation prior of SAM2 transfers well to small farmland targets. However, cockspur_grass and meadow_grass remained much lower, again confirming that thin grass-type structures are the main difficulty in this project.

To explore a fully automatic method without manual prompts, Grounded SAM2 was further tested. The method first uses Grounding DINO to detect bboxes according to the six English class names and then sends the bboxes to SAM2 for segmentation. Its AP@0.5 on the test set was only 0.2%. The main reason was that DINO could not detect extremely small targets reliably, and it also produced class-matching errors, false positives and large boxes covering almost the whole image. Therefore, the bottleneck of Grounded SAM2 in this project was the detection stage, not the subsequent SAM2 segmentation stage.

5.4 SAM2 Fine-Tuning + Classification Head

Although SAM2 zero-shot inference achieved the best segmentation result, it still depended on GT bbox prompts and could not output weed categories by itself. To further explore category prediction ability, a joint SAM2 fine-tuning + classification head strategy was designed from Week 6 to Week 8. The image encoder and prompt encoder were frozen, the mask decoder was fine-tuned, and a two-layer MLP classification head was added to predict six-class logits from 256-dimensional features after masked average pooling.

Parameter	Setting
Optimizer	AdamW, weight_decay = 1e-4
Mask decoder learning rate	3e-4
Classification head learning rate	1e-3
Loss function	mask_loss (Focal + Dice) + 0.2 x cls_loss (CrossEntropy)
Gradient accumulation	2 steps
Maximum training epochs	40 epochs, early stopping patience = 12
Experimental platform	Google Colab L4 GPU (24GB)
Training data	coco_v3 train: 2571 images / 8687 instances

Table 8. Training configuration of SAM2 fine-tuning + classification head

Two key issues were identified and fixed during training. First, the initial training loop passed original-image bbox coordinates directly into the prompt encoder, while SAM2 internally expected coordinates in 1024-space. This meant that early training was not performed correctly; the issue was fixed by using `predictor._transforms.transform_boxes()`. Second, early validation used the predicted category_id from the classification head, but the classification head had not converged, causing AP to be close to zero. After switching to GT labels to evaluate only mask quality, val AP@0.5 increased to 0.1020, indicating that the mask decoder was indeed learning.

In the final test, the model used the classification head to predict categories, and the overall AP@0.5 was only 0.1%. The fundamental reason was that the classification head did not converge stably, and category prediction errors caused COCO AP to become nearly zero. This experiment shows that joint segmentation-classification training is not suitable for the current dataset scale of 8687 instances.

Class	Test AP@0.5 with Classification Head
cockspur_grass	0.0003
black_nightshade	0.0000
field_milk_thistle	0.0034
meadow_grass	0.0002
redroot_amaranth	0.0000
white_goosefoot	0.0000
overall	0.0010

Table 9. Test results of SAM2 fine-tuning + classification head

5.5 Comparison with Related Literature

Three related works were reviewed: HQ-SAM (2306.01567), SAM2-Adapter (2408.04579) and an agricultural competition solution based on SAM (2309.13578). None of these works adopted a joint classification-head training strategy. HQ-SAM and SAM2-Adapter focus on improving segmentation quality itself and do not handle category

prediction. The agricultural SAM solution closest to this project decouples segmentation and classification completely: SAM generates masks, and an independently trained classifier processes the cropped image patches.

This comparison further supports the limitation observed in this experiment: in a small-data scenario, the joint classification-head strategy is inherently unstable. A more practical direction is to decouple SAM2 mask generation from category classification and train a separate lightweight classifier for cropped image patches.

6. Maize Row-Band Detection Experiments

6.1 Dataset and Annotation

The maize row-band detection task used the maize strip dataset provided by the internship plan. It contained 436 original images and had no existing annotations. The images were mainly near-overhead maize seedling-stage images, with strong side-light shadows, partial human shadows, sparse row bands and mixed image resolutions.

X-AnyLabeling was used to manually annotate all images with linestrip labels. Each row band was usually described with 2-3 key points: the endpoints were placed as close as possible to confirmed plant stem bases, and intermediate points were used to represent local bending. Edge vegetation clusters, isolated plants with unclear direction, and shadowed or blurred regions were not forcibly extended.

The dataset was split at an 8:1:1 ratio. The training set contained 348 original images and was expanded to 2436 images through original images plus six types of offline augmentation. The validation set contained 43 images and the test set contained 45 images; both kept the original images without augmentation. The augmentation methods included random rotation, horizontal flipping, perspective transformation and their combinations, with synchronized transformation of annotation point coordinates.

6.2 Traditional Baseline

A traditional image-processing baseline was first constructed. The pipeline included ExG vegetation extraction, column-projection peak detection, row-wise centroid sampling and RANSAC line fitting. On the test set, the detection rate was 57.6%, the mean angle error was 18.14 degrees, the mean distance error was 70.6 px, and the inference speed was about 5.5 fps.

The main problems of the traditional method were as follows. Strong shadows caused missed detections during the ExG vegetation extraction stage. Column projection assumed that row bands were close to vertical, while actual images often contained obvious tilting or local bending. Sparse row bands had insufficient foreground pixels, so the projection peaks were not obvious.

6.3 BiSeNet-V2 Semantic Segmentation and Geometric Post-Processing

The deep learning solution used BiSeNet-V2 for binary semantic segmentation, predicting the maize row-band region as a mask and then obtaining final row-band lines through geometric post-processing. BiSeNet-V2 was selected because it contains both a Detail Branch and a Semantic Branch, allowing it to combine local boundary details and global semantic information. This is suitable for thin row-band structures and also supports fast inference.

Since the original linestrip annotations needed to be converted to pixel-level labels, the polylines were dilated into fixed-width binary masks. The initial line width was set to 18, but after resizing to 512 x 256 the foreground proportion was only about 2.1%, making learning difficult. After increasing the line width to 40, the foreground proportion rose to about 6.4%, and the test IoU improved to 0.6537.

Parameter	Setting
Experimental platform	Google Colab T4 GPU
Input size	512 x 256
Batch size	8
Training epochs	60 epochs, early stopping patience = 15
Loss function	CE Loss + Dice Loss, foreground pos_weight = 5
Optimizer	SGD, momentum = 0.9, weight_decay = 1e-4
Learning rate	Poly decay, initial lr = 0.01

Table 10. BiSeNet-V2 training configuration

The post-processing pipeline was as follows. First, column projection was applied to the predicted mask to locate candidate row-band centers through peak detection. Then foreground-pixel centroids were computed row by row within each candidate region to obtain a sequence of points. RANSAC was used to fit the line $x = k \cdot y + b$. Finally, adjacent lines that were too close were merged and transformed back to original image coordinates for evaluation.

6.4 Row-Band Detection Results

Metric	Traditional Baseline	BiSeNet-V2 + Post-Processing	Change
Detection rate	57.6%	58.2%	Slight improvement
Mean angle error	18.14 degrees	2.14 degrees	Significantly reduced
Angle error < 5 degrees	18.4%	86.9%	Significantly improved
Mean distance error	70.6 px	8.6 px	Significantly reduced
Inference speed	5.5 fps	56.5 fps	Meets real-time requirement

Table 11. Comparison between BiSeNet-V2 and the traditional baseline

The results show that BiSeNet-V2 was clearly better than the traditional method in geometric accuracy and speed. The mean angle error decreased from 18.14 degrees to 2.14 degrees, the proportion of angle error below 5 degrees increased from 18.4% to 86.9%, the mean distance error decreased from 70.6 px to 8.6 px, and the complete pipeline speed reached 56.5 fps, exceeding the 20 fps real-time requirement in the internship plan.

The main remaining limitation is that the detection rate improved only slightly, from 57.6% to 58.2%. According to visual inspection, missed detections mainly occurred in two cases: first, row bands were very sparse and foreground pixels were insufficient, so column-projection peaks were not obvious; second, row bands near image edges were cropped and their line structures were incomplete. This indicates that the bottleneck is not only the segmentation model, but also the post-processing matching rules and the dataset itself.

7. Complete Experimental Comparison and Core Findings

Method	AP@0.5 / Main Metric	Prompt Required	Conclusion
SAM zero-shot (vit_l)	Overall IoU = 0.6676	GT bbox required	Strong baseline; good on broadleaf classes, weak on grass-type classes
SAM decoder-only fine-tuning	Overall IoU = 0.6289	GT bbox required	Improved some broadleaf classes but lower overall than zero-shot
Mask2Former v3	AP@0.5 = 11.6%	No	Fully automatic, but poor in extremely small target scenes
SAM2 zero-shot	AP@0.5 = 63.0%	GT bbox required	Best current scheme; four broadleaf classes exceed 90%

Method	AP@0.5 / Main Metric	Prompt Required	Conclusion
Grounded SAM2	AP@0.5 = 0.2%	No	DINO detection stage fails; difficult to handle extremely small targets
SAM2 fine-tuning + classification head	AP@0.5 = 0.1%	GT bbox required	Classification head did not converge; joint training is not feasible
BiSeNet-V2 row-band detection	Angle error = 2.14 degrees; speed = 56.5 fps	Not applicable	Geometric accuracy and speed meet the requirement; detection rate still needs improvement

Table 12. Complete comparison of experimental results

7.1 Extremely Small Targets Are the Core Limitation of End-to-End Models

Most weed instances in the current dataset are extremely small targets in 1920 x 1088 large images, with a median instance area of only about 0.05%-0.07%. When the images are resized to a short side of 480-640 for training, many target details are significantly lost. End-to-end models therefore struggle to learn stable instance location, boundary and class differences. This explains why Mask2Former did not improve significantly even after dataset expansion.

7.2 SAM2 Has Strong Segmentation Ability, but Automation Is Limited by Detection and Classification

SAM2 performed much better than end-to-end models when GT bbox prompts were available, indicating that its pretrained segmentation prior is suitable for generating small-target masks. However, practical applications still require automatic bbox detection and category prediction. The failure of Grounded SAM2 shows that current open-vocabulary detectors struggle to serve as the front-end detector in this extremely small-target farmland scenario. The SAM2 fine-tuning + classification head experiment also shows that joint category prediction is difficult under a small dataset scale.

7.3 The Deep Learning Route for Row-Band Detection Has Been Verified

For maize row-band detection, BiSeNet-V2 learned row-band region masks and significantly reduced the impact of shadows and tilted row structures compared with the traditional method. Although the detection rate has not yet reached an ideal level, the angle error, distance error and speed are all much better than the traditional baseline. This proves that semantic segmentation plus geometric post-processing is a reasonable route for the current row-band detection task.

8. Conclusions and Future Improvements

8.1 Project Summary

During the eight-week internship, a complete workflow was completed, including dataset construction, annotation, data augmentation, COCO-format conversion, model training and evaluation. The weed instance segmentation part verified the advantages of SAM/SAM2 prompt-based models for small-target segmentation and clarified the limitations of end-to-end models and open-vocabulary detectors in extremely small-target farmland scenes. The maize row-band detection part completed a full pipeline from manual linestrip annotation to BiSeNet-V2 semantic segmentation and RANSAC geometric post-processing, achieving good results in speed and geometric accuracy.

Overall, SAM2 zero-shot inference is currently the best scheme for the weed segmentation task, but it still depends on bbox prompts. Mask2Former and Grounded SAM2 do not yet meet the requirements for fully automatic high-precision segmentation. SAM2 fine-tuning + classification head joint training is not feasible under the current dataset scale. For maize row-band detection, BiSeNet-V2 basically satisfies the real-time requirement and achieves angle accuracy close to the target, but the detection rate remains the key point for future optimization.

8.2 Future Improvement Directions

- **Weed segmentation:** Prioritize a decoupled detection / segmentation / classification scheme. SAM2 can be used to generate high-quality masks, and a lightweight classifier can then be trained separately on cropped image patches, rather than training a classification head inside SAM2.
- **Grass-type classes:** For thin targets such as cockspur_grass and meadow_grass, high-resolution sliding-window inference, finer prompts or local enlarged crops can be tested to reduce the problem that targets occupy too little area inside the bbox.
- **Row-band detection:** For sparse row bands and edge-cropped samples, further optimize data augmentation, peak detection thresholds, adjacent-line merging and matching rules. Polynomial curve fitting or lane-detection representations can also be tested to improve adaptation to curved row bands.